

A reversible key uses each output letter once. One turn replaces every letter using that key; message positions stay fixed. The next turn uses the new message.

**Problem 1:** Find every reversible key that fits each clue, or explain why there is none.

**a** ABBA  $\longrightarrow$  CDDC

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**b** ABAC  $\longrightarrow$  BCBD

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**c** ABBA  $\longrightarrow$  CDCD

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**d** ABBA  $\longrightarrow$  CCCC

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**e** ABBA  $\longrightarrow$  ABBA

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**f** AABC  $\longrightarrow$  DDBA

|     |   |   |   |   |
|-----|---|---|---|---|
| in  | A | B | C | D |
| out |   |   |   |   |

**Problem 2:** Repeat each key on each message. When does the whole message first come back?

| in | A | B | C | D |
|----|---|---|---|---|
| P  | B | C | A | D |
| Q  | B | A | D | C |
| R  | B | C | D | A |

D

ABBA

ABCD

**Problem 3:** Make two different keys that change ABBA into the same message. Can their first return times for ABCD be different?

**Problem 4:** Find every possible first return time for ABCD. Make a four-letter key for each one.

**Problem 5:** Add letter E. Make a key whose first return for ABCDE is six turns. Why couldn't four letters do this?